Homework II — Solutions

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Problem 1

This problem is a nice roadmap to how a theory is canonically quantized. We start of from the classical theory defined by it's classical Lagrangian then compute all of it's quantities of interest, still classically, and lastly promote the fields to operators by imposing the canonical commutation relations, which also amounts to follow the prescription $\{\cdot,\cdot\} \to \frac{1}{\mathrm{i}\hbar}[\cdot,\cdot]$. Of course, here everything "works out" nicely, as the theory under consideration is a free theory, those are definable non-perturbatively. When we get to interacting theories not everything will function out of the box, but, many of the developments here will be useful.

1.A)

For the canonical quantization prescription, the first step is to go to the Hamiltonian phase-space formalism, in which lives the Poisson bracket, compute the conjugated momenta, and reexpress all the quantities of interest in terms of the fields and their momenta. As we know, the definition of the canonical conjugated momenta is simply the derivative of the lagrangian with respect to the time derivative of the fields,

$$\pi^I = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_I}, \quad \dot{\phi}_I = \partial_0 \phi_I \tag{1.1}$$

for our choice of Lagrangian,

$$\mathcal{L} = -\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \tag{1.2}$$

there are two kinds of dynamical objects appearing, ϕ, ϕ^*. It's clear that ϕ^* is the complex conjugate of ϕ. In principle one could argue that we have just one dynamical field, ϕ, and that ϕ^* is completely determined by ϕ. But this reasoning is not true, as a complex field, each ϕ has two real degrees of freedom, nevertheless, is not possible to extract those directly from ϕ alone. For us to define the real and imaginary parts we need a definition of ϕ^*,

$$\mathfrak{Re}[\phi] = \frac{1}{2}(\phi + \phi^*), \quad \mathfrak{Im}[\phi] = -\mathrm{i}\frac{1}{2}(\phi - \phi^*) \tag{1.3}$$

also, from complex variables calculus, we also know that z, z^* are independent quantities, which can be seen from relations of the kind,

$$\frac{\partial z^*}{\partial z} = 0 \tag{1.4}$$

here is the same principle.

All of this is just to argue that, in the expressions where we included a ϕ_I to denote multiple fields, there are two possible choices of fields, $\phi_I \in \{\phi, \phi^*\}$, that means that are two canonical conjugated momenta, one for each field[1],

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}, \quad \pi^* = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*} \tag{1.5}$$

it's straightforward to obtain then from the Lagrangian,

$$\pi = \dot{\phi}^*, \quad \pi^* = \dot{\phi} \tag{1.6}$$

[1]Actually the discussion is a lot more subtle to how the canonical momenta are defined, and which fields should have one associated with.

The next step is to proceed with the computation of the Hamiltonian density, which is given by the Legendre transform of the Lagrangian,

$$\mathcal{H} = \pi\dot{\phi} + \pi^*\dot{\phi}^* - \mathcal{L} \tag{1.7}$$
$$\mathcal{H} = \pi\pi^* + \pi^*\pi + \partial_\mu\phi^*\partial^\mu\phi + m^2\phi^*\phi \tag{1.8}$$
$$\mathcal{H} = 2\pi^*\pi - \dot{\phi}^*\dot{\phi} + \boldsymbol{\nabla}\phi^*\cdot\boldsymbol{\nabla}\phi + m^2\phi^*\phi \tag{1.9}$$
$$\mathcal{H} = 2\pi^*\pi - \pi\pi^* + \boldsymbol{\nabla}\phi^*\cdot\boldsymbol{\nabla}\phi + m^2\phi^*\phi \tag{1.10}$$
$$\mathcal{H} = \pi^*\pi + \boldsymbol{\nabla}\phi^*\cdot\boldsymbol{\nabla}\phi + m^2\phi^*\phi \tag{1.11}$$

and finally to compute the full Hamiltonian is only necessary to integrate in the spatial variables,

$$H = \int \mathrm{d}^3\mathbf{x}\,\mathcal{H} \tag{1.12}$$
$$H = \int \mathrm{d}^3\mathbf{x}\left[\pi^*\pi + \boldsymbol{\nabla}\phi^*\cdot\boldsymbol{\nabla}\phi + m^2\phi^*\phi\right] \tag{1.13}$$

1.B)

This is a nice toy model symmetry which generates a charge in the proper sense. As we'll see in due time, the electron "charge" arises due to the existence of a phase symmetry exactly as this one. As we commented before, by ϕ being a complex field, there are two degrees of freedom, this will imply in two particles in our theory, they'll have opposite sign of this charge generated by the phase symmetry. The same happens to the electron and positron.

First let's check it is a symmetry, the transformation law is,

$$\begin{cases} x & \to x' = x \\ \phi(x) & \to \phi'(x') = \mathrm{e}^{\mathrm{i}a}\phi(x) \\ \phi^*(x) & \to \phi^{*\prime}(x') = \mathrm{e}^{-\mathrm{i}a}\phi^*(x) \end{cases} \Rightarrow \begin{cases} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\Big|_{a=0} = \mathrm{i}\phi \\ \frac{\mathrm{d}\phi^{*\prime}(x)}{\mathrm{d}a}\Big|_{a=0} = -\mathrm{i}\phi \end{cases} \tag{1.14}$$

substituting back in the action,

$$S'_a[\phi'] = \int \mathrm{d}^4x'\left(-\partial'_\mu\phi^{*\prime}\partial^{\mu\prime}\phi' - m^2\phi^{*\prime}\phi'\right) \tag{1.15}$$
$$S'_a[\phi'] = \int \mathrm{d}^4x\left(-\partial_\mu\mathrm{e}^{-\mathrm{i}a}\phi^*\partial^\mu\mathrm{e}^{\mathrm{i}a}\phi - m^2\mathrm{e}^{-\mathrm{i}a}\phi^*\mathrm{e}^{\mathrm{i}a}\phi\right) \tag{1.16}$$
$$S'_a[\phi'] = \int \mathrm{d}^4x\left(-\partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi\right) = S[\phi] \tag{1.17}$$

indeed is a symmetry. Then, we can use Noether's theorem,

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi_I(x)}\frac{\mathrm{d}\phi'_I(x)}{\mathrm{d}a}\bigg|_{a=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\bigg|_{a=0} \tag{1.18}$$

as we commented before, we have to take $\phi_I = \{\phi, \phi^*\}$,

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^*(x)}\frac{\mathrm{d}\phi^{*\prime}(x)}{\mathrm{d}a}\bigg|_{a=0} \tag{1.19}$$
$$J^\mu = -\partial^\mu\phi^*\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} - \partial^\mu\phi\frac{\mathrm{d}\phi^{*\prime}(x)}{\mathrm{d}a}\bigg|_{a=0} \tag{1.20}$$
$$J^\mu = -\partial^\mu\phi^*(\mathrm{i}\phi) - \partial^\mu\phi(-\mathrm{i}\phi^*) \tag{1.21}$$

$$J^\mu = \mathrm{i}\phi^*\partial^\mu\phi - \mathrm{i}\phi\partial^\mu\phi^* \tag{1.22}$$

to obtain the conserved charge we integrate the temporal component over a spatial slice,

$$Q = \int \mathrm{d}^3\mathbf{x}\, J^0 = \int \mathrm{d}^3\mathbf{x}\left(\mathrm{i}\phi^*\partial^0\phi - \mathrm{i}\phi\partial^0\phi^*\right) \tag{1.23}$$

$$Q = -\int \mathrm{d}^3\mathbf{x}\,(\mathrm{i}\phi^*\pi^* - \mathrm{i}\phi\pi) \tag{1.24}$$

$$Q = \mathrm{i}\int \mathrm{d}^3\mathbf{x}\,(\pi\phi - \pi^*\phi^*) \tag{1.25}$$

as we'll see in the section 1.5, the minus sign between $\pi\phi$ and $\pi^*\phi^*$ is what gives a difference of sign in the charges of particles and anti-particles of this theory.

1.C)

This serves as a starting point of the quantization, showing the quantum equations of motion, of, the Heisenberg equations. As we know, these are the analogous classical equations of motion given by the Poisson bracket in the Hamiltonian formalism,

$$\{A(t,\mathbf{x}), H(t)\} = \dot{A}(t,\mathbf{x}) \Rightarrow [A(t,\mathbf{x}), H(t)] = \mathrm{i}\dot{A}(t,\mathbf{x}) \tag{1.26}$$

We already found an expression for the Hamiltonian. What is missing is the canonical commutation relations. These also mimic the canonical Poisson brackets,

applying to ϕ and substituting the already found expression for H,

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \left[\phi(t,\mathbf{x}), \int \mathrm{d}^3\mathbf{y}\left[\pi^*(t,\mathbf{y})\pi(t,\mathbf{y}) + \boldsymbol{\nabla}\phi^*(t,\mathbf{y})\cdot\boldsymbol{\nabla}\phi(t,\mathbf{y}) + m^2\phi^*(t,\mathbf{y})\phi(t,\mathbf{y})\right]\right] \tag{1.27}$$

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{y}\left[\phi(t,\mathbf{x}), \pi^*(t,\mathbf{y})\pi(t,\mathbf{y}) + \boldsymbol{\nabla}\phi^*(t,\mathbf{y})\cdot\boldsymbol{\nabla}\phi(t,\mathbf{y}) + m^2\phi^*(t,\mathbf{y})\phi(t,\mathbf{y})\right] \tag{1.28}$$

using the canonical commutation relations,

$$[\phi(t,\mathbf{x}), \phi(t,\mathbf{y})] = [\phi(t,\mathbf{x}), \phi^*(t,\mathbf{y})] = 0 \Rightarrow [\phi(t,\mathbf{x}), \boldsymbol{\nabla}\phi(t,\mathbf{y})] = [\phi(t,\mathbf{x}), \boldsymbol{\nabla}\phi^*(t,\mathbf{y})] = 0 \tag{1.29}$$

then,

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{y}\,[\phi(t,\mathbf{x}), \pi^*(t,\mathbf{y})\pi(t,\mathbf{y})] \tag{1.30}$$

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{y}\,\pi^*(t,\mathbf{y})[\phi(t,\mathbf{x}), \pi(t,\mathbf{y})] + \int \mathrm{d}^3\mathbf{y}\,[\phi(t,\mathbf{x}), \pi^*(t,\mathbf{y})]\pi(t,\mathbf{y}) \tag{1.31}$$

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{y}\,\pi^*(t,\mathbf{y})[\phi(t,\mathbf{x}), \pi(t,\mathbf{y})] \tag{1.32}$$

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{y}\,\pi^*(t,\mathbf{y})\mathrm{i}\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{1.33}$$

$$\mathrm{i}\dot{\phi}(t,\mathbf{x}) = \mathrm{i}\pi^*(t,\mathbf{x}) \Rightarrow \dot{\phi}(t,\mathbf{x}) = \pi^*(t,\mathbf{x}) \tag{1.34}$$

Analogously is possible to show that,

$$\dot{\phi^*} = \pi, \quad \dot{\pi^*} = \boldsymbol{\nabla}^2\phi - m^2\phi, \quad \dot{\pi} = \boldsymbol{\nabla}^2\phi^* - m^2\phi^* \tag{1.35}$$

1.D)

The canonical commutation relations are,

$$[\phi(t,\mathbf{x}),\pi(t,\mathbf{y})] = \mathrm{i}\delta^{(3)}(\mathbf{x}-\mathbf{y}), \quad [\phi(t,\mathbf{x}),\phi(t,\mathbf{y})] = [\pi(t,\mathbf{x}),\pi(t,\mathbf{y})] = 0 \tag{1.36}$$

$$[\phi^*(t,\mathbf{x}),\pi^*(t,\mathbf{y})] = \mathrm{i}\delta^{(3)}(\mathbf{x}-\mathbf{y}), \quad [\phi^*(t,\mathbf{x}),\phi^*(t,\mathbf{y})] = [\pi^*(t,\mathbf{x}),\pi^*(t,\mathbf{y})] = 0 \tag{1.37}$$

$$[\phi^*(t,\mathbf{x}),\phi(t,\mathbf{y})] = [\pi^*(t,\mathbf{x}),\pi(t,\mathbf{y})] = 0 \tag{1.38}$$

We expand the field in it's inverse Fourier transform,

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.39}$$

The equation of motion is,

$$\left(-\Box_x + m^2\right)\phi(x) = 0 \tag{1.40}$$

which when acting on the momentum space representation gives,

$$0 = \left(-\Box_x + m^2\right)\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\left(-\Box_x + m^2\right)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.41}$$

$$0 = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\left(p^2 + m^2\right)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.42}$$

this integral equation should be valid for every x. In fact, given an arbitrary function $\psi(x)$, we can integrate over x,

$$0\int \mathrm{d}^4x\,\psi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\left(p^2 + m^2\right)\int \mathrm{d}^4x\,\mathrm{e}^{\mathrm{i}x\cdot p}\psi(x) \tag{1.43}$$

$$0 = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\left(p^2 + m^2\right)\tilde{\psi}(-p) \tag{1.44}$$

as ψ is arbitrary, the only way to the result be zero is if $\tilde{\phi}(p)(p^2+m^2)$ is itself the zero distribution. Naively this would imply $\tilde{\phi}(p) \equiv 0$, but, due to the distributional identity $x\delta(x) = 0$, the most general solution is $\tilde{\phi}(p) \propto \delta(p^2+m^2)$. The most general solution of this equation is $\tilde{\phi}(p) = 2\pi\delta(p^2+m^2)(\theta(p^0)a(\mathbf{p}) + \theta(-p^0)b^*(\mathbf{p}))$. Thus,

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\tilde{\phi}(p)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.45}$$

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}2\pi\delta\left(p^2+m^2\right)\left(\theta\left(p^0\right)a(\mathbf{p}) + \theta\left(-p^0\right)b^*(\mathbf{p})\right)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.46}$$

to simplify this we need to use some additional properties of the Dirac delta,

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^3 2\omega_{\mathbf{p}}}\left(\delta\left(p^0-\omega_{\mathbf{p}}\right) + \delta\left(p^0+\omega_{\mathbf{p}}\right)\right)\left(\theta\left(p^0\right)a(\mathbf{p}) + \theta\left(-p^0\right)b^*(\mathbf{p})\right)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.47}$$

$$\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^3 2\omega_{\mathbf{p}}}\left(\delta\left(p^0-\omega_{\mathbf{p}}\right)a(\mathbf{p}) + \delta\left(p^0+\omega_{\mathbf{p}}\right)b^*(\mathbf{p})\right)\mathrm{e}^{\mathrm{i}x\cdot p} \tag{1.48}$$

$$\phi(x) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}}\left(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + b^*(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\right) \tag{1.49}$$

Now we use the commutation relations of the fields to obtain the commutation relations of the creation add annihilation operators,

1.E)

Let's start with the Hamiltonian,

$$H = \int \mathrm{d}^3\mathbf{x} \left[\pi^*\pi + \boldsymbol{\nabla}\phi^* \cdot \boldsymbol{\nabla}\phi + m^2\phi^*\phi\right] \tag{1.50}$$

$$\begin{aligned} H = \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{k}}{(2\pi)^6 4\omega_\mathbf{p}\omega_\mathbf{k}} &\Big[\partial_0\big(a(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} + b^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big)\partial_0\big(b(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + a^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big) \\ &+ \boldsymbol{\nabla}\big(b(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} + a^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big) \cdot \boldsymbol{\nabla}\big(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big) \\ &+m^2\big(b(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} + a^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big)\big(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big)\Big] \end{aligned} \tag{1.51}$$

$$\begin{aligned} H = \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{k}}{(2\pi)^6 4\omega_\mathbf{p}\omega_\mathbf{k}} &\Big[\mathrm{i}\omega_\mathbf{k}\big(-a(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} + b^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big)\mathrm{i}\omega_\mathbf{p}\big(-b(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + a^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big) \\ &+ \mathrm{i}^2\mathbf{p}\cdot\mathbf{k}\big(b(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} - a^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big)\big(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} - b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big) \\ &+m^2\big(b(\mathbf{k})\mathrm{e}^{\mathrm{i}x\cdot k} + a^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}x\cdot k}\big)\big(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big)\Big] \end{aligned} \tag{1.52}$$

$$\begin{aligned} H = \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{k}}{(2\pi)^6 4\omega_\mathbf{p}\omega_\mathbf{k}} &\Big[\mathrm{e}^{\mathrm{i}x\cdot(k+p)}\big(-\omega_\mathbf{k}\omega_\mathbf{p}a(\mathbf{k})b(\mathbf{p}) - \mathbf{k}\cdot\mathbf{p}b(\mathbf{k})a(\mathbf{p}) + m^2b(\mathbf{k})a(\mathbf{p})\big) \\ &+ \mathrm{e}^{\mathrm{i}x\cdot(k-p)}\big(\omega_\mathbf{k}\omega_\mathbf{p}a(\mathbf{k})a^\dagger(\mathbf{p}) + \mathbf{k}\cdot\mathbf{p}b(\mathbf{k})b^\dagger(\mathbf{p}) + m^2b(\mathbf{k})b^\dagger(\mathbf{p})\big) \\ &+ \mathrm{e}^{\mathrm{i}x\cdot(-k+p)}\big(\omega_\mathbf{k}\omega_\mathbf{p}b^\dagger(\mathbf{k})b(\mathbf{p}) + \mathbf{k}\cdot\mathbf{p}a^\dagger(\mathbf{k})a(\mathbf{p}) + m^2a^\dagger(\mathbf{k})a(\mathbf{p})\big) \\ &+\mathrm{e}^{-\mathrm{i}x\cdot(k+p)}\big(-\omega_\mathbf{k}\omega_\mathbf{p}b^\dagger(\mathbf{k})a^\dagger(\mathbf{p}) - \mathbf{k}\cdot\mathbf{p}a^\dagger(\mathbf{k})b^\dagger(\mathbf{p}) + m^2a^\dagger(\mathbf{k})b^\dagger(\mathbf{p})\big)\Big] \end{aligned} \tag{1.53}$$

$$\begin{aligned} H = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 4\omega_\mathbf{p}^2} &\Big[\mathrm{e}^{-\mathrm{i}x^0 2\omega_\mathbf{p}}\big(-\omega_\mathbf{p}\omega_\mathbf{p}a(-\mathbf{p})b(\mathbf{p}) + \mathbf{p}^2b(-\mathbf{p})a(\mathbf{p}) + m^2b(-\mathbf{p})a(\mathbf{p})\big) \\ &+ \big(\omega_\mathbf{p}\omega_\mathbf{p}a(\mathbf{p})a^\dagger(\mathbf{p}) + \mathbf{p}^2b(\mathbf{p})b^\dagger(\mathbf{p}) + m^2b(\mathbf{p})b^\dagger(\mathbf{p})\big) \\ &+ \big(\omega_\mathbf{p}\omega_\mathbf{p}b^\dagger(\mathbf{p})b(\mathbf{p}) + \mathbf{p}^2a^\dagger(\mathbf{p})a(\mathbf{p}) + m^2a^\dagger(\mathbf{p})a(\mathbf{p})\big) \\ &+\mathrm{e}^{\mathrm{i}x^0 2\omega_\mathbf{p}}\big(-\omega_\mathbf{p}\omega_\mathbf{p}b^\dagger(-\mathbf{p})a^\dagger(\mathbf{p}) + \mathbf{p}^2a^\dagger(-\mathbf{p})b^\dagger(\mathbf{p}) + m^2a^\dagger(-\mathbf{p})b^\dagger(\mathbf{p})\big)\Big] \end{aligned} \tag{1.54}$$

$$H = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 4\omega_\mathbf{p}^2}\left[\omega_\mathbf{p}^2a(\mathbf{p})a^\dagger(\mathbf{p}) + \omega_\mathbf{p}^2b(\mathbf{p})b^\dagger(\mathbf{p}) + \omega_\mathbf{p}^2b^\dagger(\mathbf{p})b(\mathbf{p}) + \omega_\mathbf{p}^2a^\dagger(\mathbf{p})a(\mathbf{p})\right] \tag{1.55}$$

$$H = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 4\omega_\mathbf{p}}\omega_\mathbf{p}\left[2(2\pi)^3 2\omega_\mathbf{p}\delta^{(3)}(0) + 2b^\dagger(\mathbf{p})b(\mathbf{p}) + 2a^\dagger(\mathbf{p})a(\mathbf{p})\right] \tag{1.56}$$

$$H = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_\mathbf{p}}\omega_\mathbf{p}\left[b^\dagger(\mathbf{p})b(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p})\right] + 2\delta^{(3)}(0)\int \mathrm{d}^3\mathbf{p}\,\frac{1}{2}\omega_\mathbf{p} \tag{1.57}$$

1.F)

$$\Delta_\mathrm{F}(x,y) = \langle 0|\mathrm{T}\big\{\phi(x)\phi^\dagger(y)\big\}|0\rangle \tag{1.58}$$

$$\Delta_\mathrm{F}(x,y) = \theta\big(x^0 - y^0\big)\langle 0|\phi(x)\phi^\dagger(y)|0\rangle + \theta\big(y^0 - x^0\big)\langle 0|\phi^\dagger(y)\phi(x)|0\rangle \tag{1.59}$$

$$\Delta_\mathrm{F}(x,y) = \int \frac{\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{k}}{(2\pi)^6 4\omega_\mathbf{p}\omega_\mathbf{k}}\Big(\theta\big(x^0 - y^0\big)\langle 0|a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p}a^\dagger(\mathbf{k})\mathrm{e}^{-\mathrm{i}y\cdot k}|0\rangle + \theta\big(y^0 - x^0\big)\langle 0|b(\mathbf{k})\mathrm{e}^{\mathrm{i}y\cdot k}b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}|0\rangle\Big) \tag{1.60}$$

$$\Delta_{\mathrm{F}}(x,y) = \int \frac{\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} \Big(\theta\big(x^0-y^0\big)\mathrm{e}^{-\mathrm{i}y\cdot k+\mathrm{i}x\cdot p}(2\pi)^3 2\omega_{\mathbf{p}}\delta^{(3)}(\mathbf{k}-\mathbf{p})\,\langle 0|0\rangle$$
$$+\theta\big(y^0-x^0\big)\mathrm{e}^{-\mathrm{i}x\cdot p+\mathrm{i}y\cdot k}(2\pi)^3 2\omega_{\mathbf{p}}\delta^{(3)}(\mathbf{k}-\mathbf{p})\,\langle 0|0\rangle\Big) \tag{1.61}$$
$$\Delta_{\mathrm{F}}(x,y) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} \Big(\theta\big(x^0-y^0\big)\mathrm{e}^{\mathrm{i}(x-y)\cdot p} + \theta\big(y^0-x^0\big)\mathrm{e}^{-\mathrm{i}(x-y)\cdot p}\Big) \tag{1.62}$$
$$\Delta_{\mathrm{F}}(x,y) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}}\mathrm{e}^{\mathrm{i}(\mathbf{x}-\mathbf{y})\cdot\mathbf{p}} \Big(\theta\big(x^0-y^0\big)\mathrm{e}^{-\mathrm{i}(x^0-y^0)\omega_{\mathbf{p}}} + \theta\big(y^0-x^0\big)\mathrm{e}^{\mathrm{i}(x^0-y^0)\omega_{\mathbf{p}}}\Big) \tag{1.63}$$

Let us use the integral representation of the Heaviside function,

$$\theta(x) = -\frac{1}{2\pi\mathrm{i}}\lim_{\epsilon\to 0^+}\int_{-\infty}^{+\infty}\mathrm{d}p\,\frac{\mathrm{e}^{-\mathrm{i}xp}}{p+\mathrm{i}\epsilon} \tag{1.64}$$

then,

$$\Delta_{\mathrm{F}}(x,y) = -\frac{1}{2\pi\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}}\mathrm{e}^{\mathrm{i}(\mathbf{x}-\mathbf{y})\cdot\mathbf{p}}\int \mathrm{d}p^0\,\frac{1}{p^0+\mathrm{i}\epsilon}\Big(\mathrm{e}^{-\mathrm{i}(x^0-y^0)p^0}\mathrm{e}^{-\mathrm{i}(x^0-y^0)\omega_{\mathbf{p}}} + \mathrm{e}^{\mathrm{i}(x^0-y^0)p^0}\mathrm{e}^{\mathrm{i}(x^0-y^0)\omega_{\mathbf{p}}}\Big) \tag{1.65}$$

shifting the p^0 integral,

$$\Delta_{\mathrm{F}}(x,y) = -\frac{1}{2\pi\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}}\mathrm{e}^{\mathrm{i}(\mathbf{x}-\mathbf{y})\cdot\mathbf{p}}\int \mathrm{d}p^0\,\frac{1}{p^0-\omega_{\mathbf{p}}+\mathrm{i}\epsilon}\Big(\mathrm{e}^{-\mathrm{i}(x^0-y^0)p^0} + \mathrm{e}^{\mathrm{i}(x^0-y^0)p^0}\Big) \tag{1.66}$$

changing the sign of p^0 in the second term,

$$\Delta_{\mathrm{F}}(x,y) = -\frac{1}{2\pi\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}}\mathrm{e}^{\mathrm{i}(\mathbf{x}-\mathbf{y})\cdot\mathbf{p}}\int \mathrm{d}p^0\,\mathrm{e}^{-\mathrm{i}(x^0-y^0)p^0}\left(\frac{1}{p^0-\omega_{\mathbf{p}}+\mathrm{i}\epsilon} + \frac{1}{-p^0-\omega_{\mathbf{p}}+\mathrm{i}\epsilon}\right) \tag{1.67}$$
$$\Delta_{\mathrm{F}}(x,y) = -\frac{1}{\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^4 p}{(2\pi)^4 2\omega_{\mathbf{p}}}\mathrm{e}^{\mathrm{i}(x-y)\cdot p}\left(\frac{-2\omega_{\mathbf{p}}+2\mathrm{i}\epsilon}{(p^0-\omega_{\mathbf{p}}+\mathrm{i}\epsilon)(-p^0-\omega_{\mathbf{p}}+\mathrm{i}\epsilon)}\right) \tag{1.68}$$
$$\Delta_{\mathrm{F}}(x,y) = \frac{1}{\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^4 p}{(2\pi)^4}\frac{\mathrm{e}^{\mathrm{i}(x-y)\cdot p}}{-(p^0)^2+\omega_{\mathbf{p}}^2-2\mathrm{i}\epsilon\omega_{\mathbf{p}}-\epsilon^2} \tag{1.69}$$
$$\Delta_{\mathrm{F}}(x,y) = \frac{1}{\mathrm{i}}\lim_{\epsilon\to 0^+}\int \frac{\mathrm{d}^4 p}{(2\pi)^4}\frac{\mathrm{e}^{\mathrm{i}(x-y)\cdot p}}{p^2+m^2-\mathrm{i}\epsilon} \tag{1.70}$$

A Conventions

A.1 Metric

There are two choices of metric,

$$g_{-} = \mathrm{diag}\big(- \quad + \quad + \quad +\big), \quad g_{+} = \mathrm{diag}\big(+ \quad - \quad - \quad -\big) \tag{A.1}$$

I will always use g_{-}. This changes a lot of signs in a lot of places, as example,

$$\mathcal{L} = g_{+}^{\mu\nu}\frac{1}{2}\partial_\mu\phi^*\partial_\nu\phi - V[\phi,\phi^*] = -g_{-}^{\mu\nu}\frac{1}{2}\partial_\mu\phi^*\partial_\nu\phi - V[\phi,\phi^*] \tag{A.2}$$

Physics should be invariant under conventions, so they don't matter, as long as you know which conventions you are using and are consistent.

A.2 Levi-Civita

The Levi-Civita is the (unique) totally antisymmetric four-pseudo-tensor, $\epsilon_{\alpha\beta\mu\nu}$, there are two possible choices of normalization,

$$\epsilon^{+}_{0123} = 1, \quad \epsilon^{-}_{0123} = -1 \tag{A.3}$$

I will always use ϵ^{+}. We define also the spatial Levi-Civita by restricting the covariant one,

$$\epsilon_{ijk} \coloneqq \epsilon_{0ijk} \tag{A.4}$$

A.3 Creation and annihilation operators

When expanding the field in normal modes, there is a choice of normalization of the integral,

$$\phi(x) = \int \frac{\mathrm{d}^3\mathbf{p}}{N_{\mathbf{p}}}\big(a(\mathbf{p})\mathrm{e}^{\mathrm{i}x\cdot p} + b^\dagger(\mathbf{p})\mathrm{e}^{-\mathrm{i}x\cdot p}\big) \tag{A.5}$$

some common choices are,

$$N_{\mathbf{p}} = \begin{cases} (2\pi)^3 2\omega_{\mathbf{p}} \\ (2\pi)^3\sqrt{2\omega_{\mathbf{p}}} \\ \sqrt{(2\pi)^3 2\omega_{\mathbf{p}}} \end{cases} \tag{A.6}$$

this changes the commutation relation of the operators as,

$$\big[a(\mathbf{p}), a^\dagger(\mathbf{k})\big] = \frac{N_{\mathbf{p}}^2}{(2\pi)^3 2\omega_{\mathbf{p}}}\delta^{(3)}(\mathbf{p}-\mathbf{k}) \tag{A.7}$$

and also the definition of a momentum eigenstate,

$$|\mathbf{p}\rangle = \frac{(2\pi)^3 2\omega_{\mathbf{p}}}{N_{\mathbf{p}}}a^\dagger(\mathbf{p})\,|0\rangle\,, \quad \langle\mathbf{k}|\mathbf{p}\rangle = 2\omega_{\mathbf{p}}(2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}) \tag{A.8}$$

I will always use $N_{\mathbf{p}} = (2\pi)^3 2\omega_{\mathbf{p}}$.